MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DECODER, BCD TO DECIMAL, DIGITAL, KEY CODE BMH

This amendment forms a part of Military Specification MIL-M-28787/258A(EC), dated 14 August 1981, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4 AND 5

TABLE I: Delete and substitute new table I as printed on pages 2 and 3 of this amendment.

PAGE 6

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

6.4, second line: Delete "100°C" and substitute "85°C".

PAGE 7

6.4, Output voltage (V_{DD} = 5 V): Delete "100°C" (two places) and substitute "85°C" (two places).

6.4, Input voltage (all inputs): Delete "100°C" (two places) and substitute "85°C" (two places).

PAGE 16

FIGURE 6: Delete and substitute new figure 6 as printed on page 4 of this amendment.

PAGE 22

FIGURE 8: Delete and substitute new figure 8 as printed on page 5 of this amendment.

PAGE 24

FIGURE 9: Delete and substitute new figure 9 as printed on page 6 of this amendment.

PAGE 27

FIGURE 10: Delete and substitute new figure 10 as printed on page 7 of this amendment.

PAGE 31

FIGURE 11: Delete and substitute new figure 11 as printed on page 8 of this amendment.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N499)

FSC 5963

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A6	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1.0	∞	1.0	∞	1.0	∞	1.0	∞	1.0	∞	MΩ
C1-C8	Negative input clamp diode voltage	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
D1-D8	Positive input clamp diode voltage	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
E1-E10	Functional	VI	X	← See table VI →										
F1-F20	Static output voltage, low level	VII	X	0.0	0.05	0.0	0.05	0.0	0.05	0.0	0.05	0.0	0.05	V
G1-G20	Static output voltage, high level	VIII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
H1-H20	Static output voltage, low level	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J20	Static output voltage, high level	X	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
K1-K20	Static output voltage, low level	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
L1-L20	Static output voltage, high level	XII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
M1-M8	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
N1-N8	Input current, low level	XIV	X	0	-200	---	---	0	-200	---	---	0	-200	nA
P	Static power supply current	XV	X	0	40	0	40	0	580	0	40	0	968	µA
Q1-Q20	Transition time (t_{TLH})	XVI	S	35	350	35	350	38	385	26	265	41	413	ns
Q21-Q40	Transition time (t_{TLH})	XVI	Q	83	830	83	830	91	915	63	630	97	978	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R20	Transition time (t_{THL})	XVII	S	17	175	17	175	19	195	13	135	20	208	ns
R21-R40	Transition time (t_{THL})	XVII	Q	65	655	65	655	72	725	50	500	77	775	ns
S1-S60	Propagation delay time (t_{PLH})	XVIII	S	48	480	48	480	53	530	36	365	57	568	ns
S61-S120	Propagation delay time (t_{PLH})	XVIII	Q	68	680	68	680	75	750	51	515	80	803	ns
T1-T60	Propagation delay time (t_{PHL})	XIX	S	48	480	48	480	53	530	36	365	57	568	ns
T61-T120	Propagation delay time (t_{PHL})	XIX	Q	68	680	68	680	75	750	51	515	80	803	ns
U1-U20	Transition time (t_{TLH})	XX	Q	15	150	15	150	16	165	11	115	17	178	ns
V1-V20	Transition time (t_{THL})	XXI	Q	7	75	7	75	8	85	5	55	9	88	ns
W1-W60	Propagation delay time (t_{PLH})	XXII	Q	20	200	20	200	22	220	15	150	23	236	ns
X1-X60	Propagation delay time (t_{PHL})	XXIII	Q	20	200	20	200	22	220	15	150	23	236	ns
Y1-Y20	Transition time (t_{TLH})	XXIV	Q	11	110	11	110	12	120	8	85	13	129	ns
Z1-Z20	Transition time (t_{THL})	XXV	Q	5	55	5	55	6	60	4	42	6	64	ns
AA1-AA60	Propagation delay time (t_{PLH})	XXVI	Q	14	140	14	140	15	155	10	105	16	164	ns
BB1-BB60	Propagation delay time (t_{PHL})	XXVII	Q	14	140	14	140	15	155	10	105	16	164	ns
CC1-CC4	Capacitor	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
DD	Power supply filter capacitance	XXIX	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
EE1-EE8	Input capacitance	XXX	Q	2	15	2	15	2	15	2	15	2	15	pF

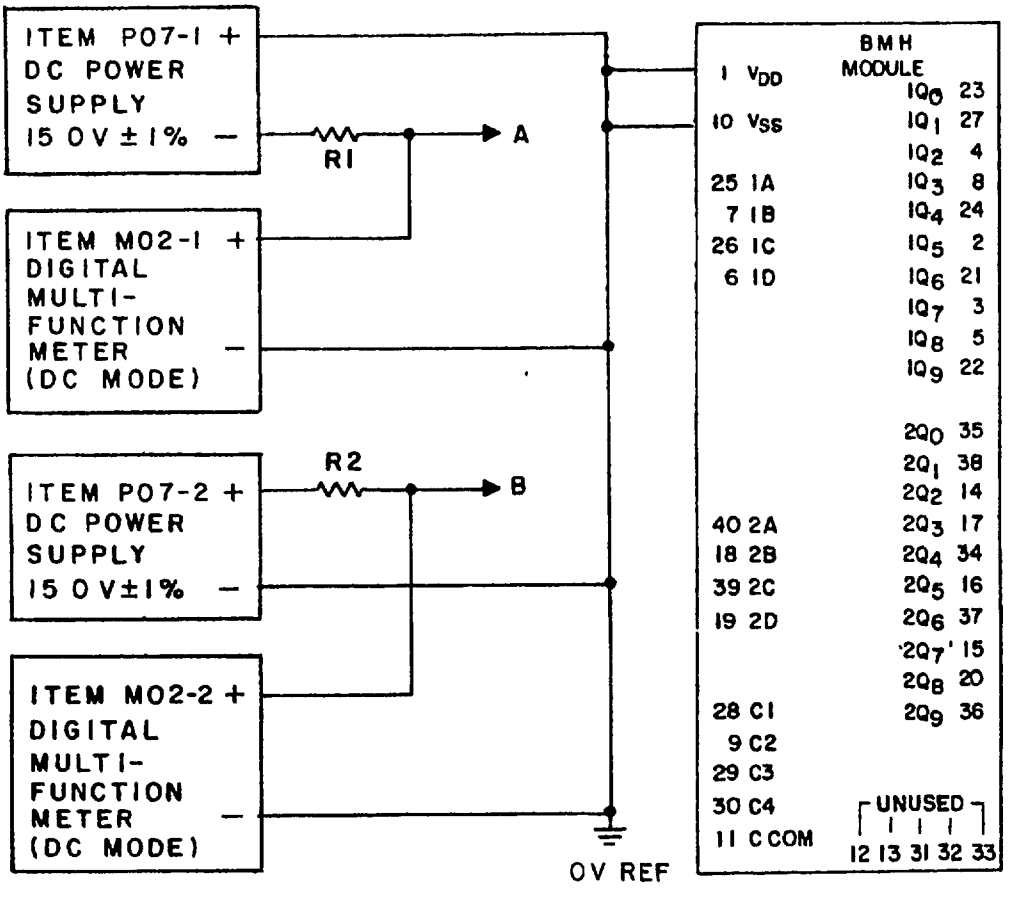
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified limits are for initial and end-of-life.

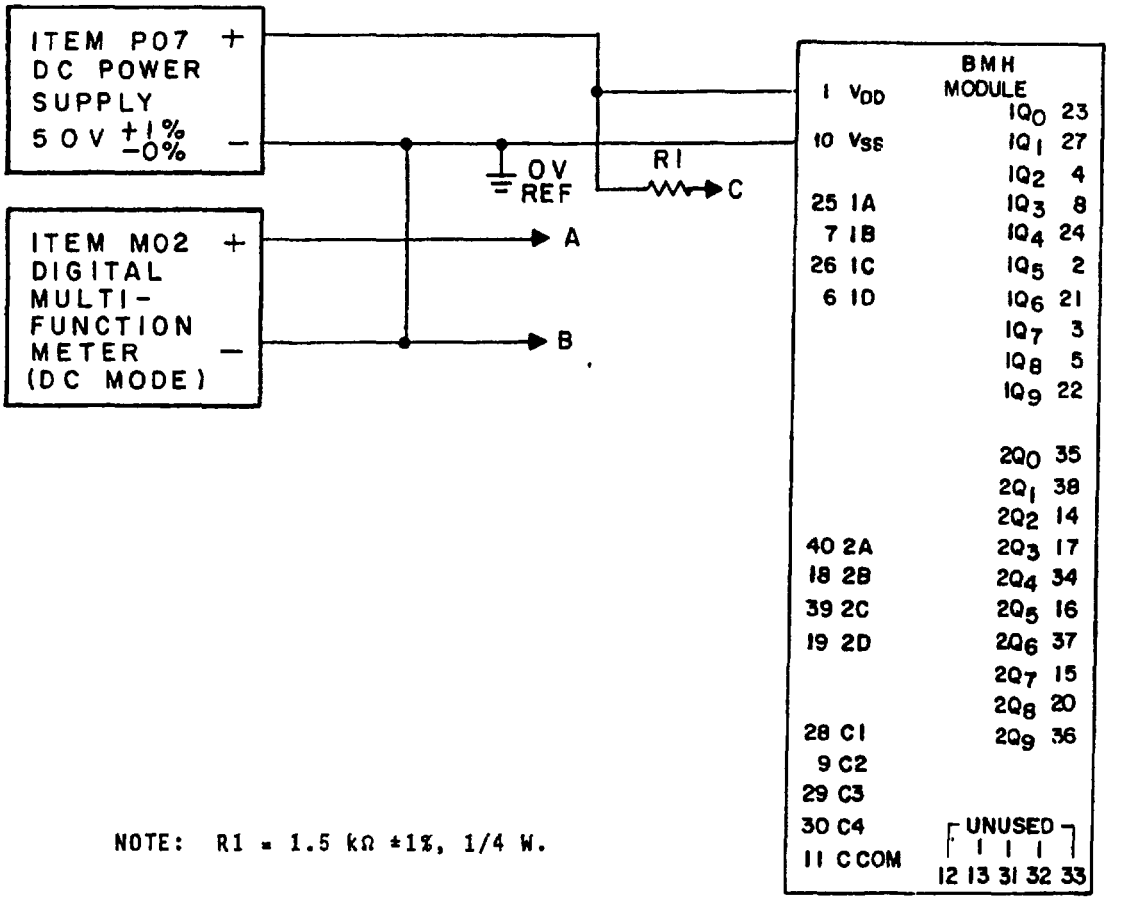
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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Only item P07 and one item M02 is required for test conditions C and D.

FIGURE 6. Positive and negative input clamp diode voltage test circuit for tables IV and V.

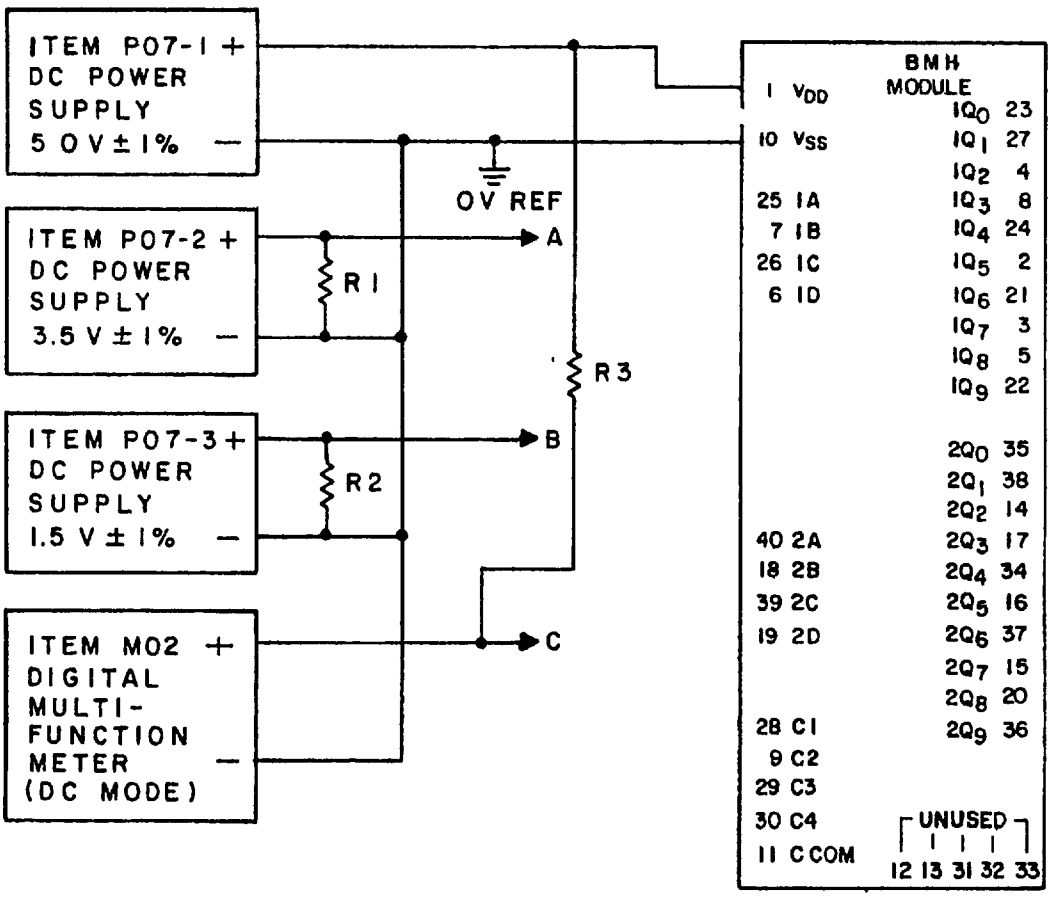
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NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 8. Static output voltage (low-level and high-level) test circuit for tables VII and VIII.

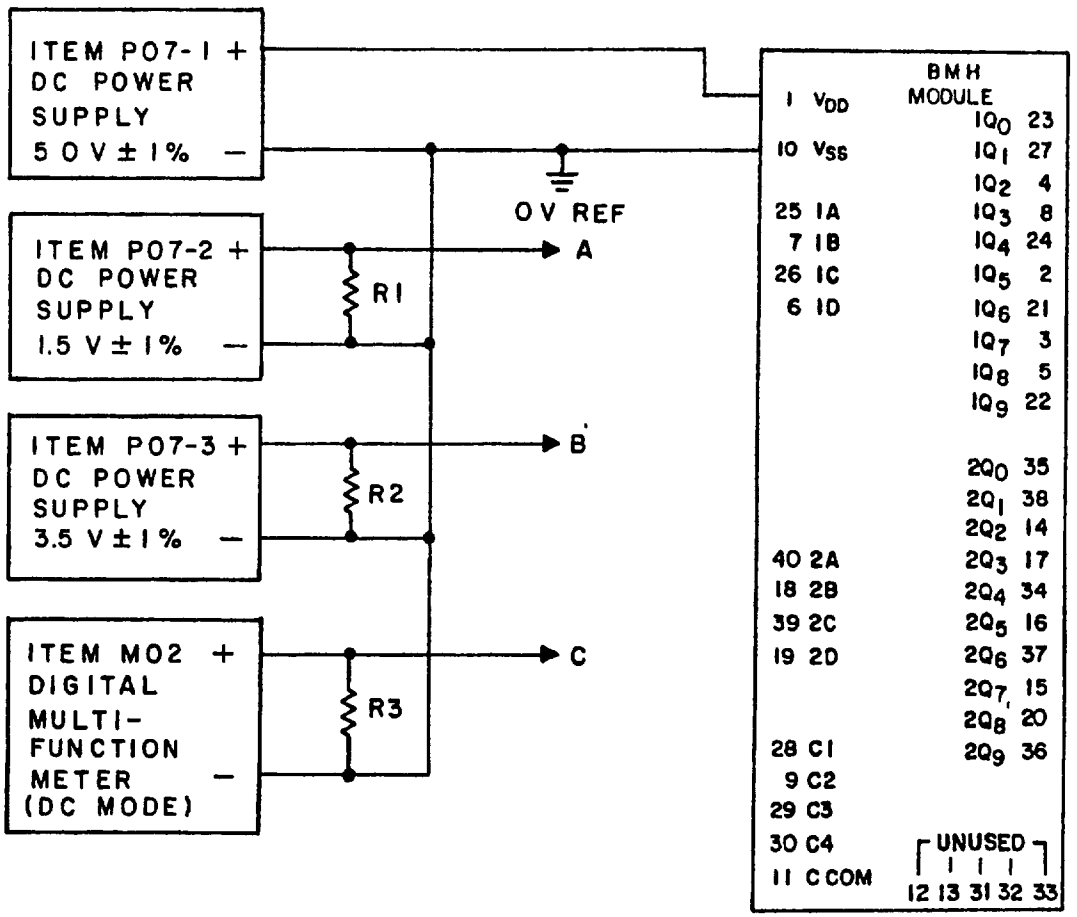
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 7.15 kΩ ±1%, 1/8 W @ -55°C.
= 9.09 kΩ ±1%, 1/8 W @ 0°C and 25°C.
= 10.5 kΩ ±1%, 1/8 W @ 60°C.
= 12.7 kΩ ±1%, 1/8 W @ 85°C.

FIGURE 9. Static output voltage (low-level) test circuit for table IX.

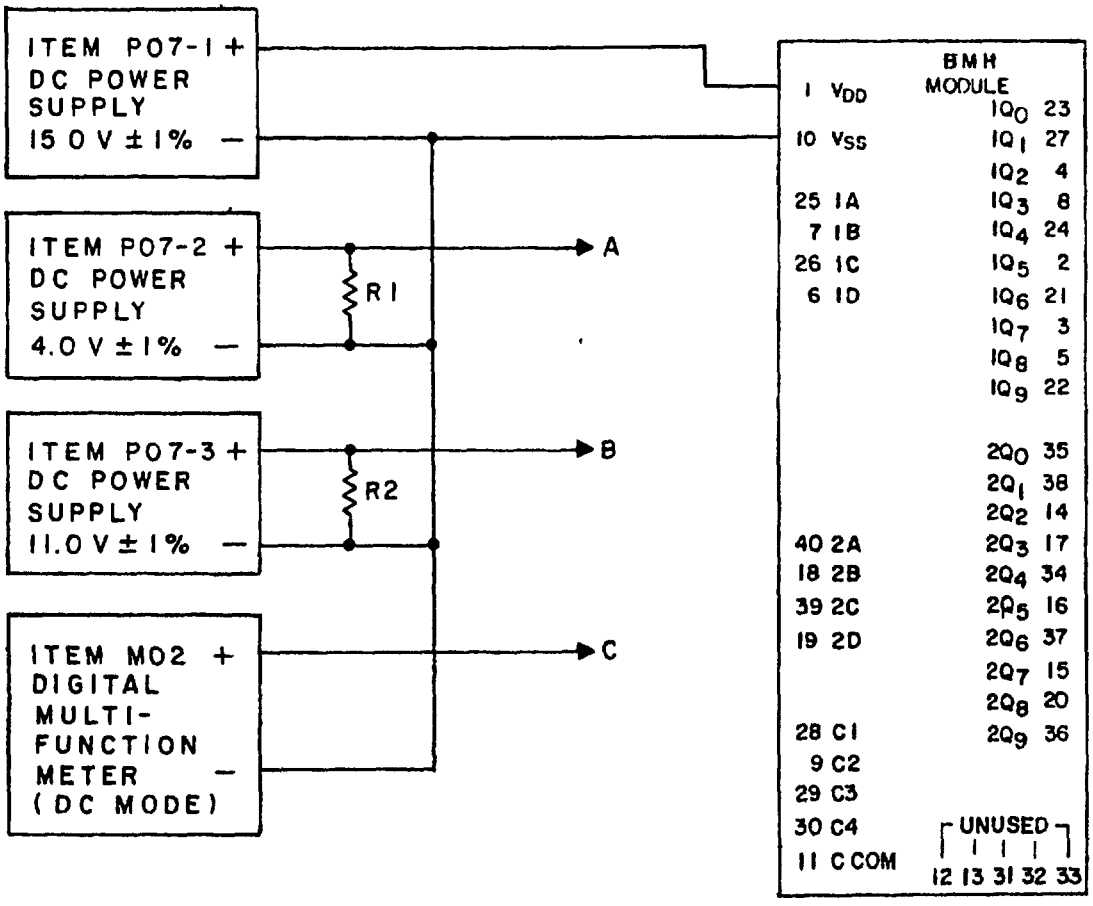
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 18.5 kΩ ±1%, 1/8 W @ -55°C.
= 23.2 kΩ ±1%, 1/8 W @ 0°C and 25°C.
= 27.4 kΩ ±1%, 1/8 W @ 60°C.
= 32.4 kΩ ±1%, 1/8 W @ 85°C.

FIGURE 10. Static output voltage (low-level) test circuit for table X.

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/2 W.

FIGURE 11. Static output voltage (low-level and high-level) test circuit for tables XI and XII.

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